



Fig. 6: Ready to use



Fig. 9: No propeller detected



Fig. 7: Test passed successfully



Fig. 10: No power detected



Fig. 8: Motor is overloaded



Fig. 11: No motor detected

For construction, follow the wiring diagram shown in Fig. 5 to connect all components along with the Arduino Nano expansion board. Begin by mounting the Nano and expansion board on a stable base or inside a suitable enclosure. Secure the ESC and BLDC motor firmly to minimise vibration during testing.

Connect the three-phase wires of the BLDC motor to the ESC terminals (A, B, and C). Next, connect the LiPo battery to the ESC power

input (red wire to positive and black wire to ground), ensuring proper polarity and insulation for all high-current connections. The ESC signal wire connects to a PWM-capable digital pin of the Arduino, and the ESC ground connects to the common ground of the Arduino to ensure proper operation.

Next, connect the sensing and display modules. The INA219 current sensor connects in series with the battery's positive line to measure current, and its SDA and SCL pins connect to the Arduino's

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I²C pins (A4 and A5 in the Uno/Nano). The voltage sensor module connects across the battery terminals, and its analogue output connects to an Arduino analogue input pin such as A0. The OLED display also interfaces through I²C (SDA and SCL) and VCC and GND connections.

The RGB LED connects to three digital pins through appropriate current-limiting resistors, while the push button connects to a digital input pin using either a pull-up or a pull-down resistor. Finally, connect the buzzer module to a digital output pin.

After verifying all connections according to the circuit diagram and ensuring a common ground for all modules, power the system and test the setup. The prototype is now ready for testing, as shown in Fig. 1.

For testing, power the system and the ESC using a separate LiPo battery. Observe the RGB LED status under different device conditions as listed in Table 2. The OLED display shows the corresponding messages. Refer to Figs 6, 7, 8, 9, 10, and 11 for the display outputs under various conditions. **EFY**

Neel Soni is a robotics engineer with over seven years of hands-on experience in robotics, drones, and embedded systems, specialising in practical systems built using Arduino, ESP32, and other microcontrollers. He has earned 50+ certificates from national and international robotics competitions and technical events and has trained numerous students through hands-on robotics and embedded systems workshops.

